Department of Mathematics and Applied Mathematics

3MS Metric Spaces

Tutorial 11 **2024**

1. The metric space $B(X)$ with the supremum metric has proved to be a very useful complete metric space for us. One might wonder about whether it is compact. Prove: For any non-empty set X, $B(X)$ with the supremum metric is not compact, because it is not bounded.

2. **Theorem** If $f : (X, d_X) \to (Y, d_Y)$ is a continuous function between metric spaces and (X, d_X) is compact, then f is uniformly continuous.

   Proof Suppose that f is continuous, but not uniformly continuous. (1)
   Then there exists $\epsilon > 0$ and sequences (x_n) and (y_n) such that, for all $n \in \mathbb{N}$, $d_X(x_n, y_n) < \frac{1}{n}$ but $d_Y(f(x_n), f(y_n)) \geq \epsilon$. (2)
   Then $d_X(x_n, y_n) \to 0$ as $n \to \infty$. (3)
   Let (x_{n_k}) be a convergent subsequence of (x_n); say $x_{n_k} \to x$ as $k \to \infty$. (4)
   Then $y_{n_k} \to x$ also. (5)
   Then $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$. (6)
   So $d_Y(f(x_{n_k}), f(y_{n_k})) \to 0$. (7)
   This contradicts the statement in line (2). (8)

   (a) Write down the definition of uniform continuity (using ϵ's and δ's); negate it and use that to explain where line (2) comes from.
   (b) How do we know that the subsequence (x_{n_k}) mentioned in line (4) exists?
   (c) Why does $y_{n_k} \to x$ as claimed in line (5)?
   (d) Why does $f(x_{n_k}) \to f(x)$ as claimed in line (6)?
   (e) Explain the contradiction mentioned in line (8).

3. (a) Consider the following incorrect proof (supposedly from an actual textbook) that any totally bounded set S in a metric space is bounded. Give an example of a totally bounded set in the reals with the usual metric, to illustrate where the proof breaks down.

   Incorrect argument The set S, being totally bounded, can be covered by finitely many balls of radius 1, say by N balls of radius 1. Then S is a subset of any ball $B(x, 2N)$ of radius $2N$ provided that the centre x lies in S. Thus diam $S \leq 4N$ so that S is bounded.

   (b) Provide a correct proof that every totally bounded subset of a metric space is bounded.

4. We say that a family $\mathcal{F}$ of sets has the **finite intersection property** if the intersection of any finite subfamily of $\mathcal{F}$ is non-empty. (It's not particularly good terminology, but it is used in most texts.) Rewrite the definition of a space having compact topology in terms of closed sets, in such a way that you use the term "finite intersection property."
This shouldn't take long; I just want you to be able to recognize the notion of "compact topology" when you see it written in terms of closed sets, rather than open ones.

• The proof below is optional. The instructions are merely: fill in any further details you need to understand this proof.

5. **Theorem** A metric space is complete and totally bounded if and only if it has compact topology.

Proof ($\Longrightarrow$) Use proof by contradiction. So assume X is complete and totally bounded, but does not have compact topology.
Then there is an open cover $\mathcal{U}$ of X that has no finite subcover.
Since X is totally bounded, there exists a finite $F \subseteq X$ such that

$$X = \bigcup \left\{ B(x; \frac{1}{2}) \ : \ x \in F \right\}.$$

At least one of these sets, say $G_1 = B(x; \frac{1}{2})$ for some x, cannot be covered by finitely many members of $\mathcal{U}$.
Note that diam $G_1 \leq 1$.

Now G_1 is totally bounded, so there exists a finite $F_1 \subseteq G_1$ such that

$$G_1 = \bigcup \left\{ B(x; \frac{1}{4}) \ : \ x \in F_1 \right\}$$

(where the open ball is taken in the subspace G_1).
At least one of these sets, say $G_2 = B(x; \frac{1}{4})$ for some x, cannot be covered by finitely many members of $\mathcal{U}$.
Note that diam $G_2 \leq \frac{1}{2}$.

Continue in this way, to obtain a decreasing sequence (G_n) of non-empty subsets of X with diam $G_n \leq \frac{1}{n} \to 0$.
Now use completeness: by Cantor's Intersection Theorem, $\bigcap \overline{G}_n \neq \emptyset$.

Take $z \in \bigcap \overline{G}_n$.
Since $\mathcal{U}$ is a cover of X, there exists $U \in \mathcal{U}$ such that $z \in U$.
Since U is open, there exists $r > 0$ such that $B(z; r) \subseteq U$.
Take n such that diam $\overline{G}_n =$ diam $G_n \leq \frac{1}{n} < r$.
Then $\overline{G}_n \subseteq B(z; r) \subseteq U$.
So G_n is covered by $\{U\}$, a subcover of $\mathcal{U}$ with just *one* member - contradiction.
($\Longleftarrow$) We omit.